WHITE PAPER · SUPPLY RESILIENCE

Protecting Continuity When Supply Is Constrained

A criticality-led approach to supply resilience, and what leaders can do before the next disruption

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> **Based on first-hand experience.** This paper draws on my own experience of leading the supply response for a global industrial technology company through the component shortage that followed the COVID-19 pandemic.

SUMMARY

When a global shortage of components reached finished products, customers waited and the business felt it directly. Recovery required more than expediting. It needed a single accountable view of continuity, a way of deciding what mattered most, a set of levers matched to specific risks, and a rhythm that aligned every function behind shared decisions. This paper describes that approach, the trade-offs it forced into the open, how the results should be read honestly, and what leaders can build before disruption tests their own supply chains.

01

When a shortage reaches the customer

The disruption that followed the pandemic was, for many manufacturers, a lesson in how far upstream risk travels. Lead times for electronic components rose sharply, and finished-product lead times followed them, in some cases to many times their normal level. Customers who had been used to short, predictable delivery were left waiting, and the business was exposed on the products that mattered most to its revenue and reputation.

Six constraints made the problem harder than the shortage alone.

- **Limited visibility.** Available inventory and the dates of next supply were often uncertain, which meant surprises late in the chain.
- **A tension between cost and continuity.** Procurement had to hold its cost targets, while the business needed a view of what was critical to keep shipping. Each concern was legitimate. What was missing was one place where they were weighed together.
- **Scarce supply with competing claims.** Where supply was constrained, it had to be prioritised across customers and orders, and delivery dates had to be communicated clearly.
- **No single owner of continuity.** The responsibility sat across several functions, each looking at its own part of the chain.
- **Supply redirected to higher bidders.** Component suppliers sometimes redirected available supply to other customers willing to pay a premium. Waiting times then grew without warning, and the delays were unplanned rather than forecast.
- **Open-market premiums.** Some critical components were bought in bulk by brokers and resold on the open market at a premium. That pushed up procurement cost and left the business choosing between paying more and waiting longer.

> *The shortage was a supply problem. The recovery was a governance problem.*

02

Give continuity an owner

The first decision was organisational. The business gave one role a management mandate over delivery prioritisation and customer commitments, with an obligation on the functions involved to involve that role. The role was accountable for the customer receiving material as committed, and it covered the whole chain from demand projection to the delivery date promised to the customer.

That role had no team and no reporting line over the functions it depended on. This is an important design point. Authority came from a mandate, and results came from a shared view of the risk, a steady cadence and decisions that everyone could see. It also meant that the leader could not afford to be distant. Regular, close contact with the plant kept the picture honest, from frequent visits to a daily presence once the role became full time on site.

Leaders who face a similar situation should ask whether continuity has a named owner, and whether that owner has the authority to make calls that other functions must respect.

03

A criticality lens

The most useful reframe was a simple one. Instead of asking how to speed up everything, we asked what would stop shipments, and concentrated effort there.

The approach works in layers. First, rank the product lines by where a stoppage would hurt customers and revenue most. Second, within the most critical line, rank the products by demand, function and customer impact. Third, take the components of those products and place them on a nine-box grid, classified by market supply conditions, whether they are custom or standard, how critical they are, and whether alternates exist. That exercise singled out a small set of very critical components from among hundreds.

Each of those components then received a decision rather than a general intention: secure an alternate, build a buffer, or dual-source, with several of these running in parallel.

The lens did two things. It added a continuity view alongside cost discipline, and it anchored every decision to the product roadmap, so that effort was not wasted on products nearing the end of their life.

04

A portfolio of levers

No single fix brought lead times down. The recovery came from several levers working together, each matched to the risk it removes.

LEVER	RISK IT REMOVES	HOW IT WAS APPLIED
Alternates	Long or uncertain lead times	Proposed alternates were qualified with engineering, written into the bill of materials, and purchasing began according to availability.
Buffer stock	Components with no alternate	Sized on cost, market availability, quality and the impact on the business.
Dual sourcing	Single-source exposure	A second supplier identified, samples tested by engineering, and cost and reliability reviewed.
Premium purchases	Parts unavailable through normal channels	Recommended from a criticality view, with the business impact set out for each, and approved by finance.
Product roadmap	End-of-life and paired products	Replacement timing used to plan supply, and products that ship together planned as a unit.

The same generic approach applied across the product families, with intensity and focus varying according to importance. The product line that mattered most received the most attention, and it led the recovery.

05

Governing the trade-offs

A crisis forces trade-offs into the open. The test of good governance is whether each one is made explicitly, with an owner, a business case and an approval.

Cost against continuity

Paying a premium for scarce parts is sometimes the right answer. It is a decision, not a reflex. Each such purchase was recommended from a supply-criticality view, with its business impact set out, and finance made the call. That structure kept the spending defensible, and it kept the business from paying for certainty it did not need.

Supplier relationships

Supplier relationships come under strain when supply is scarce. Visits to validate a supplier's set-up, quality and committed timelines were more valuable than remote reassurance. Where supply was held up by commercial matters rather than by capacity, it was escalated and resolved through the right approvals.

Customer priority

Where there is not enough to go round, someone has to decide who is served first. Weekly reviews with customers, customer-facing teams and sales produced a clear call on priority, and clear, communicated delivery dates followed from it. Customers value a reliable date they can plan around more than an optimistic one that moves.

> *Every trade-off was made explicit, with an owner, a business case and an approval.*

06

The operating rhythm

Resilience is sustained by rhythm. The recovery depended on a small number of routines that kept the whole enterprise looking at the same picture.

- **A weekly crisis forum** of senior leaders from finance, procurement, R&D, production and sales, so that decisions were fast, informed and shared.
- **A weekly report** everyone could rely on: status, new challenges, items resolved and pending, impacted orders and customers, and a rolling action list.
- **Challenge, not acceptance.** Demand projections, production plans, buffer levels and obsolescence risk were reviewed every cycle, and tested against history and against what sales, engineering and the factory knew.
- **Customer communication** that gave sales teams and customers clear visibility of delivery dates.

None of these routines is exotic. Their value came from consistency, and from the fact that a senior leader from every relevant function was in the room.

07

Reading the results honestly

By the time of my departure from the role, lead times had fallen by 60% or more across the major product families, led by the most important product line. On-time delivery had improved materially, and it did so while the shortage was still under way. Customers felt significant relief, and the recovery was regarded across the business as a major achievement.

It is worth reading these results with care. Component supply never fully normalised. Lead times on the lead product were still above their pre-disruption baseline, and the final stretch was paced deliberately, balancing the benefit of shorter lead times against the cost of holding inventory. A recovery described honestly is more credible than one described as complete.

08

What leaders can take from this

THREE LESSONS

- **Give continuity an owner.** Put the question of what is critical alongside cost discipline, with one accountable view.
- **Invest in visibility early.** Better systems to track the chain and the availability of material would have reduced the surprises. This is the investment I would make first.
- **Pay for certainty deliberately.** A premium is a decision with a business case, not a reflex.

The deeper lesson is about timing. Much of what made recovery possible, such as the criticality view, the alternates qualified with engineering, the dual-sourcing relationships and the forum itself, is far cheaper to build before a shortage than during one. Resilience is cheapest when the pressure is lowest.

09

Questions to ask before the next disruption

A SHORT CHECKLIST FOR LEADERSHIP TEAMS

- Which products and components could stop shipments, and does the business agree on the list?
- Where there is no alternate, is there a buffer, and who decided its size?
- Where there is a single source, what would it take to qualify a second?
- Who owns continuity, and does that person have authority across functions?
- How quickly can the business see inventory and the date of next supply?
- Is there a standing forum with senior representation from finance, procurement, engineering, production and sales?
- What is the process for approving a premium purchase, and who approves it?
- How does the product roadmap inform supply decisions, particularly for products nearing the end of their life?

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Where I can help

I work with leadership teams on three areas:

- **Supply criticality assessment.** Which products and components can stop revenue, and what to do about each.
- **Shortage-recovery governance.** A cross-functional forum, a reporting rhythm and clear decision rights.
- **Resilience design.** Alternates, buffer and dual sourcing, tied to the product roadmap.

The most valuable moment for this work is before a shortage forces the issue, but it is also useful in the middle of one.

About the author. Raghunandan Gopalan has spent three decades across engineering, R&D, PLM and supply chain, on both the services side and the client side. At Schneider Electric he led the move of a core engineering capability from a partner-run model to a captive Global Capability Center in India, and led the supply response that restored delivery of Process Automation products through the global component shortage. He advises leadership teams on GCC strategy and set-up, and on supply resilience.

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